

Saheed Faremi

Data scientist & founding engineer · PhD researcher, deep learning for EEG

Production ML & LLM systems, owned end to end

saheedfaremi@gmail.com · Based in Dublin, Ireland



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[Semantic Scholar](#)

Data scientist and founding engineer shipping production ML systems in fintech and agriculture since 2021, and a PhD researcher in deep learning for EEG at University College Cork (Artificial Intelligence and Cognitive Load Research Lab, supervised by Luca Longo). Built the fraud/AML intelligence layer for a payments platform operating in eight African markets, and shipped LLM document automation that cut manual work by 85%. First-author research published in Brain Informatics and IEEE, with code released open-source. UNESCO India-Africa Hackathon 2022 gold medallist.

EXPERIENCE

PhD researcher · Artificial Intelligence and Cognitive Load Research Lab (AICL), University College Cork

Present

Doctoral research on segmenting EEG into microstate sequences with variational autoencoders, supervised by Luca Longo. Builds generative models, large-scale evaluation sweeps, and temporal post-hoc analyses over resting-state EEG.

- Developed Conv-VaDE, a convolutional variational autoencoder with a learnable Gaussian-mixture latent prior and four polarity-invariance mechanisms, for segmenting EEG topographic maps into microstate sequences
- Ran a 4,832-model architecture search (486 configurations per participant across state count, latent dimension, depth, and width), scoring 110 metrics per model with cross-subject ICC consistency analysis on SLURM GPU and IBM Power9 clusters
- Developed Graph-VaDE, a graph-neural-network VAE with a learnable Gaussian-mixture prior that discovers microstates directly on the electrode graph via static anatomical adjacency, replacing interpolated topographic images; recovers ground-truth maps on a synthetic gate benchmark (ARI 1.0, GEV 0.85)
- Built a post-hoc hidden semi-Markov backfitting stage that learns per-state dwell-time distributions and transition probabilities, then relabels full recordings coherently

- Quantified recurrence in resting EEG (single-participant case study): the multichannel signal recurs well above phase-randomised surrogates (determinism 0.83 against 0.60) while the discrete label sequence does not exceed a first-order Markov surrogate

Data scientist → Founding engineer · Etihuku (Curnance)

2021-05 → present

Data science and engineering at Etihuku, a data analytics consultancy, including founding-engineer ownership of Curnance, its multi-asset fintech venture.

- Founding engineer for Curnance, the multi-asset fintech venture built within Etihuku: shipped seven production services covering wallet/ledger, auth, tiered KYC/KYB, a Go jobs handler, a Flutter mobile app (~120 screens), a SvelteKit admin console, and a Next.js site
- Closed a live withdrawal-fraud race condition by moving every external payout path to debit-before-pay ordering, with row-level locking and deterministic idempotency keys
- Designed a rules-first, LLM-last fraud/AML intelligence layer: 38 deterministic detectors, LLM-drafted suspicious-activity reports, and a compliance Q&A analyst guarded by k-anonymity floors and PII redaction
- Built an end-to-end document-generation system with large language models on Azure ML Studio, cutting manual creation time by 85% while meeting compliance requirements across three regions
- Deployed ML models via FastAPI on Microsoft Azure, serving predictive analytics for agriculture (500+ farmers), financial inclusion (90% classification accuracy), and compliance systems
- Automated build and deployment with per-service Azure DevOps pipelines and Bicep infrastructure, reducing time to production by 40%; quality gates include 680+ automated tests and formal VAPT remediation

PUBLICATIONS

Peer-reviewed

Saheed Faremi, Luca Longo "Integrating Convolutional Variational Autoencoders and the Gaussian Mixture Model for efficient manifold learning and clustering of spatially preserved EEG topographic maps". *Brain Informatics*, 2026. [doi:10.1186/s40708-026-00327-9](https://doi.org/10.1186/s40708-026-00327-9)

Akinpelumi Saheed Faremi, Boluwaji Akinnuwesi, Elliot Mbunge, Petros M. Mashwama, Stephen Fashoto, Polite Zenzo Ncube, John Batani, Shamsudeen Ademola Sanni, Yinusa A. Faremi, Andile Metfula "Machine Learning Models for Identifying Factors Influencing and Predicting Malaria Among Children Under Five Years in Nigeria". *IEEE ICTAS 2024*, 2024. [doi:10.1109/ICTAS59620.2024.10507142](https://doi.org/10.1109/ICTAS59620.2024.10507142)

Preprints, registered protocols & doctoral work

Morteza Akbari, **Saheed Akinpelumi Faremi**, Luca Longo "Autoencoder-Based Models for Scalp EEG: A Systematic Review of Architectures, Applications, Latent Representations, Interpretability, and Validation". *OSF Registries (registered systematic-review protocol)*, 2026. [doi:10.17605/osf.io/v2w3z](https://doi.org/10.17605/osf.io/v2w3z)

Saheed Faremi, Andrea Visentin, Luca Longo "Interpretable EEG Microstate Discovery via Variational Deep Embedding: A Systematic Architecture Search with Multi-Quadrant Evaluation". *XAI 2026 (Late-breaking work + Doctoral Consortium track), Fortaleza, Brazil (arXiv preprint)*, 2026. [doi:10.48550/arXiv.2605.10947](https://doi.org/10.48550/arXiv.2605.10947) [code](#)

Saheed Faremi "Explainable Disentangled Representation Learning of Recurring Brain Activation Patterns via Variational Autoencoders". *XAI World Conference 2025, Doctoral Proposals track*, 2025.

EDUCATION

PhD · University College Cork

In progress

Neural engineering and machine learning
Supervisor: Luca Longo

MSc · Technological University Dublin

Machine learning, deep learning, data mining, visualisation, statistics
Taught master's in computer science with a machine-learning concentration; coursework in deep learning, data mining, statistical modelling, and visualisation. Completed 2025 with a 4.0/4.0 GPA (80/100).

BSc · University of Eswatini

Information technology
Bachelor of Science in Information Technology, awarded with Second Class Honours, Upper Division (2.1).

TECHNICAL SKILLS

AI & MACHINE LEARNING	PyTorch · TensorFlow · scikit-learn · CatBoost · LLMs (Azure OpenAI, OpenAI, Anthropic) · RAG · Fraud and anomaly detection · Transformer fine-tuning (BERT, RoBERTa) · Explainable AI
PROGRAMMING	Python · SQL · Go · TypeScript · Bash
CLOUD & MLOPS	Azure (ML Studio, AI Studio, Functions, DevOps, Bicep) · CI/CD (GitHub Actions, Azure DevOps) · SLURM cluster computing
DATA & SYSTEMS	ETL/ELT pipelines · MySQL · Double-entry ledger systems · RESTful APIs and microservices
STATISTICS & ANALYTICS	Statistical hypothesis testing (Wilcoxon, ICC, multiple-comparison correction)

RECOGNITION

Winner, UNESCO India-Africa Hackathon 2022 (AGRI12) 2022

UNESCO · Ministry of Education Innovation Cell (India) · Gold medal + ₹3 lakh team prize

Deep Learning IndabaX: Churn Prediction Challenge 2022

Deep Learning IndabaX · 3rd of 21

Google NLP Hack Series: Swahili Sentiment Analysis 2023

Google NLP Hack Series · 4th of 29

UmojaHack Africa 2023: Cryptojacking Detection 2023

UmojaHack Africa · 35th of 328 (top 11%)

SELECTED PROJECTS

AI-assisted Farmer Call Center

Automated voice-response system letting farmers report issues via phone, SMS, or web and receive AI-routed answers. · Team engineer, Eswatini representative

Python · Voice AI · Twilio

EEG microstate analysis with variational autoencoders

Source segmentation of EEG signals via variational autoencoders, including a GMM-VAE for soft clustering. · PhD researcher

Python · PyTorch · MNE · NumPy · scikit-learn

Multilingual sentiment analysis for Swahili

Transformer models fine-tuned for sentiment classification of Swahili tweets; placed 4th of 29 in the Google NLP Hack Series. · Author

Python · BERT · RoBERTa

TALKS

Interpretable EEG Microstate Discovery via Variational Deep Embedding · XAI 2026 (Late-breaking work + Doctoral Consortium track), 2026

Machine Learning Models for Predicting Malaria in Nigerian Children Under Five · IEEE ICTAS 2024, 2024